Mendelian Susceptibility to Mycobacterial Disease due to Partial IRF8 Deficiency — Comprehensive Disease Report

MONDO:0013957 · OMIM 614893 (IMD32A) · Orphanet 319600 · DOID:0111986 · MedGen 814919 (UMLS C3808589) · GARD 0017463

Summary

Mendelian Susceptibility to Mycobacterial Disease due to **partial IRF8 deficiency** (MSMD-IRF8, MONDO:0013957) is a rare **autosomal dominant** inborn error of immunity caused by a **heterozygous, dosage-sensitive** variant in *IRF8* (Interferon Regulatory Factor 8), a master myeloid transcription factor located at chromosome 16q24.1. The prototypical allele, **p.Thr80Ala (T80A; c.238A>G)**, lies within the IRF8 DNA-binding domain and disrupts IRF8–DNA binding, thereby reducing IRF8 transcriptional activity. Because myeloid cell fate is determined in an IRF8 **dose-dependent** manner, a partial (heterozygous) defect produces a **graded, subset-selective** immune phenotype: a *selective depletion of circulating CD11c⁺CD1c⁺ conventional dendritic cells (cDC2)*, while sparing monocytes and other DC subsets. This is fundamentally milder than the autosomal recessive **complete** IRF8 deficiency (IMD32B, OMIM 614894; e.g., p.Lys108Glu/K108E), in which monocytes and all dendritic cells are absent and a life-threatening syndrome ensues.

Clinically, partial IRF8 deficiency presents in childhood as **curable disseminated BCG disease** or environmental/non-tuberculous mycobacterial infection, reflecting a functional bottleneck in the **IL-12/23 → IFN-γ circuit** that all ~22 genetic etiologies of MSMD share. The two originally described T80A subjects (Hambleton et al., NEJM 2011) were otherwise healthy and their mycobacterial disease was curable, in stark contrast to the recessive K108E patient, who required hematopoietic stem-cell transplantation. The mechanistic basis — impaired

dendritic-cell antigen presentation and IL-12 production feeding into reduced IFN-γ and defective macrophage activation against intracellular mycobacteria — anchors the entire disease definition.

This report consolidates database-verified identifiers, variant classifications (ClinVar: T80A "likely pathogenic," K108E "pathogenic," both absent from gnomAD v4), UniProt-confirmed protein mapping (both residues fall within the IRF tryptophan pentad repeat DNA-binding domain, aa 7–114), authoritative HPO phenotype annotations sourced to

P 21524210 ,

and the orthologous BXH2 mouse model, into a single knowledge-base entry spanning all 15 requested sections.

Key Findings

Finding 1 — Partial IRF8 deficiency (autosomal dominant T80A) causes MSMD via selective cDC2 depletion

The landmark study of Hambleton and colleagues (PMID: 21524210, *NEJM* 2011) identified two distinct *IRF8* mutations defining two distinct diseases. The **T80A** variant was heterozygous and produced an **autosomal dominant, milder immunodeficiency** with a **selective depletion of CD11c$^+$CD1c$^+$ circulating dendritic cells** — the entity captured by MONDO:0013957. The authors state verbatim: *"The T80A variant was associated with an autosomal dominant, milder immunodeficiency and a selective depletion of CD11c$^+$CD1c$^+$ circulating dendritic cells."* They further note that they *"studied two otherwise healthy subjects with a history of disseminated but curable BCG disease in childhood,"* establishing both the clinical phenotype (curable disseminated BCG disease) and the small denominator (n=2) that underlies the disease's phenotype frequencies. Mechanistically, *"Both K108E and T80A mutations impair IRF8 transcriptional activity by disrupting the interaction between IRF8 and DNA."* This finding is the central pillar of the report: partial IRF8 deficiency = dose-sensitive cDC2 loss driving curable MSMD.

Finding 2 — All MSMD etiologies converge on the IL-12/23–IFN-γ circuit

Partial IRF8 deficiency is one member of a larger genetic family. A systematic review of **830 MSMD patients from 581 families** (PMID: 38341181, Khavandegar et al. 2024) catalogued 299 unique mutations across 21 genes, with **lymphadenopathy the most common manifestation (378/830, 45.5%; multifocal in 35.1%)**, followed by fever (30.2%), organomegaly (24.8%), and sepsis (20.8%). The mean age was 10.4 years, the highest patient frequencies were in Iran, Turkey, and Saudi Arabia, and 45.5% had a positive family history. A 2026 review (PMID: 42183200, Qian et al.) states that *"22 genes have been implicated, all converging on the IL-12/23-IFN-γ circuit, underscoring its non-redundant role in controlling intracellular pathogens,"* with roughly 50% of patients still molecularly unsolved. This establishes the shared pathophysiologic funnel into which IRF8 deficiency feeds: impaired dendritic cell / macrophage cytokine cross-talk that cripples IFN-γ–dependent control of mycobacteria.

Finding 3 — IRF8 is a dose-dependent master transcription factor of the DC lineage; validated by the BXH2 mouse

The dose-sensitivity that explains why heterozygous T80A gives a subset-selective (rather than global) defect is well documented. Nishiyama & Tamura 2025 (PMID: 40680811) show that IRF8 is pivotal for type 1 conventional DC (cDC1) differentiation, establishes the enhancer landscape at the progenitor stage, and that *"the cell fate within the myeloid lineages is determined in an IRF8 dose-dependent manner."* Bigley et al. 2018 (PMID: 29128673) anchor the model organism: the human IRF8 **R291Q** variant is *"orthologous to R294, which is mutated in the BXH2 IRF8-deficient mouse,"* and IRF8 mutants *"failed to regulate the Ets/IRF composite element (EICE) or interferon-stimulated response element (ISRE)."* Ham et al. 2025 (PMID: 40072380) confirm that a **dominant-negative** IRF8 form causes decreased cDC2 and mycobacterial susceptibility, phenotypically distinct from the recessive severe form. Together these establish the graded genotype→cell-fate→phenotype logic.

Finding 4 — T80A and K108E are database-classified pathogenic variants absent from population databases

Database verification (gnomAD API, GRCh38) confirms *IRF8* = ENSG00000140968, chr16:85,899,162–85,922,606, canonical transcript ENST00000268638. Via ClinVar: **c.238A>G p.Thr80Ala** (16-85909053-A-G) is classified **"Likely pathogenic"** (missense); **c.322A>G p.Lys108Glu** (16-85909137-A-G) is classified **"Pathogenic"** (missense). **Neither variant appears in gnomAD v4** (absent from >730,000 population alleles), consistent with ultra-rare, high-penetrance disease alleles. The original functional evidence underlying these classifications is Hambleton 2011's demonstration that both mutations *"impair IRF8 transcriptional activity by disrupting the interaction between IRF8 and DNA."*

Finding 5 — Corrected identifiers and protein-domain mapping

EBI OLS4 confirms MONDO:0013957 = "Mendelian susceptibility to mycobacterial diseases due to partial IRF8 deficiency," cross-referenced to **OMIM:614893, Orphanet:319600, DOID:0111986, MedGen:814919 (UMLS C3808589), GARD:0017463**, with synonyms including "immunodeficiency 32A / IMD32A" and "autosomal dominant … partial deficiency." The **partial/dominant form** = **OMIM 614893** = **IMD32A** (correcting an earlier A/B label swap); the **complete/recessive form** = **OMIM 614894** = **IMD32B**. UniProt **Q02556** (IRF8, 426 aa) shows residue 80 = Thr, 108 = Lys, 83 = Arg; the sole annotated DNA-binding feature — the **"IRF tryptophan pentad repeat" spanning aa 7–114** — contains both Thr80 and Lys108, explaining why both variants disrupt IRF8–DNA interaction.

Finding 6 — Official HPO phenotype annotations for OMIM:614893

The JAX HPO annotation network for OMIM:614893 (gene NCBIGene:3394 *IRF8*), all sourced to PMID: 21524210, lists: **HP:0020086 BCGitis (2/2)**, **HP:0032252 Granuloma (2/2)**, **HP:0002716 Lymphadenopathy (2/2)**, **HP:0011463 Childhood onset (2/2)**, **HP:0001945 Fever (1/2)**, **HP:0002840 Lymphadenitis (1/2)**, HP:0002721 Immunodeficiency, HP:0002719 Recurrent infections, and **HP:0000006 Autosomal dominant inheritance**. Frequencies derive from the two otherwise-healthy T80A subjects with curable disseminated BCG disease.

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. A **heterozygous missense variant in *IRF8*** (prototypically c.238A>G, p.Thr80Ala) arises in the germline **→ results in** a single mutant IRF8 allele encoding a protein with an amino-acid substitution in the DNA-binding domain (IRF tryptophan pentad repeat, aa 7–114).
2. The Thr80Ala substitution **→ disrupts the IRF8–DNA interaction** (demonstrated in vitro; Hambleton 2011), lowering IRF8's ability to bind composite EICE/ISRE elements **→ results in** reduced IRF8 transcriptional output (a partial, not complete, loss of function).

3. Because myeloid lineage commitment is **IRF8 dose-dependent** (Nishiyama & Tamura 2025), the reduced functional IRF8 dose → **leads to** a selective failure to specify/maintain the **CD11c⁺CD1c⁺ conventional dendritic cell (cDC2)** compartment, while monocytes and other subsets are relatively spared (contrast: complete deficiency ablates all DCs + monocytes).
4. Selective cDC2 depletion → **impairs** antigen presentation and, critically, **IL-12/IL-23 production** by the dendritic-cell compartment (*inferred* from the shared MSMD circuit; the DC subset is a physiological IL-12 source feeding the axis).
5. Reduced IL-12/23 → **results in** blunted **IFN-γ** production by T cells and NK cells (the non-redundant MSMD circuit; Qian 2026).
6. Deficient IFN-γ signaling → **leads to** inadequate **macrophage activation** and failure to kill ingested intracellular mycobacteria.
7. Uncontrolled mycobacterial replication (BCG vaccine strain, environmental/non-tuberculous mycobacteria, or *M. tuberculosis*) → **produces** the clinical phenotype: **granuloma formation, lymphadenitis/lymphadenopathy, BCGitis, fever**, and disseminated but (in the partial form) **curable** mycobacterial disease.

Branch point: The same gene, when hit by a *biallelic complete* loss-of-function (recessive K108E) or a *dominant-negative* allele (e.g., c.1279dupT p.*427Leuext*42), diverts to a **more severe branch** — loss of monocytes and all DC subsets (K108E) or additional pDC/cDC1 loss with broadened viral susceptibility (dominant-negative) — producing a life-threatening syndrome that typically requires HSCT.

```
       IRF8 T80A (het, DBD)                  IRF8 K108E (biallelic)
                |                                      |
   partial ↓ transcriptional activity      complete loss of function
                |                                      |
    selective cDC2 (CD1c+) depletion       loss of ALL DCs + monocytes
                |                                      |
       ↓ IL-12/23 from DC                     global APC failure
                |                                      |
           ↓ IFN-γ                           severe multi-lineage defect
                |                                      |
  ↓ macrophage killing of mycobacteria       overwhelming infection
                |                                      |
  CURABLE disseminated BCG/NTM disease       LIFE-THREATENING → HSCT
  (MSMD, IMD32A / OMIM 614893)               (IMD32B / OMIM 614894)
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Coverage of mechanism checklist

- **Molecular pathways:** IL-12/23 → IFN-γ signaling axis (JAK-STAT/STAT1 downstream); IRF8 transcriptional regulation via EICE (Ets/IRF composite) and ISRE elements. GO terms: **GO:0035722** (interleukin-12-mediated signaling), **GO:0060333** (interferon-gamma-mediated signaling), **GO:0006357** (regulation of transcription by RNA Pol II), **GO:0002250** (adaptive immune response).
- **Cellular processes:** dendritic cell differentiation (**GO:0097028**), myeloid cell differentiation (**GO:0030099**), antigen processing and presentation, granuloma formation, macrophage activation (**GO:0042116**).
- **Protein dysfunction:** loss of function via disrupted DNA binding (partial, T80A) vs. loss of nuclear localization/stability (K108E; P 25122610) vs. dominant-negative sequestration (c.1279dupT; P 40072380).
- **Immune system involvement:** immunodeficiency (predisposition to intracellular pathogens); not primarily autoimmune, though granulomatous/inflammatory manifestations occur.
- **Cell types (CL):** conventional dendritic cell type 2 / CD1c⁺ DC (**CL:0001057 / CL:0002399**), conventional dendritic cell (**CL:0000990**), monocyte (**CL:0000576**), macrophage (**CL:0000235**), plasmacytoid DC (**CL:0000784**), neutrophil (**CL:0000775**).

Anatomical, Temporal, Population, Diagnostic, Treatment & Related Sections

Anatomical structures affected

- **Primary:** immune/hematopoietic system — bone marrow myeloid progenitors, circulating dendritic cell compartment (**UBERON:0002371** bone marrow; **UBERON:0000178** blood).
- **Secondary organ involvement:** lymph nodes (**UBERON:0000029**) — lymphadenopathy/ lymphadenitis; spleen and liver (**UBERON:0002106 / UBERON:0002107**) — organomegaly; skin — BCGitis at inoculation site; potentially disseminated (lungs, bone).

- **Subcellular:** nucleus (**GO:0005634**) — site of IRF8 transcriptional action; the T80A defect impairs nuclear DNA binding, the K108E defect impairs nuclear import.
- **Lateralization:** typically regional/multifocal lymphadenopathy (35.1% multifocal in the MSMD cohort), not strictly lateralized.

Temporal development

- **Onset:** childhood (HP:0011463); mean age across MSMD ~10.4 years; BCG complications appear after neonatal/infant vaccination in BCG-vaccinating countries.
- **Course:** infection-triggered and episodic; the partial form is comparatively mild and **curable** with antimycobacterial therapy — distinct from the chronic/lethal recessive form.
- **Critical period:** post-BCG vaccination in infancy; environmental mycobacterial exposure throughout childhood.

Inheritance and population

- **Inheritance:** autosomal dominant (HP:0000006), dosage-sensitive haploinsufficiency / partial loss of function.
- **Penetrance/expressivity:** variable; the disease is defined from very few families, so precise penetrance is unknown. Both T80A carriers were otherwise healthy apart from mycobacterial disease.
- **Allele frequency:** T80A and K108E both absent from gnomAD v4 (ultra-rare).
- **Epidemiology:** MSMD collectively is rare; highest reported patient frequencies in Iran, Turkey, Saudi Arabia (partly reflecting consanguinity for recessive forms and endemic TB/BCG use). Partial IRF8 deficiency specifically is described in only a handful of families.

Diagnostics

- **Immunophenotyping (flow cytometry):** selective reduction of circulating **$CD11c^+CD1c^+$ (cDC2)** dendritic cells with preserved monocytes — the hallmark distinguishing partial from complete IRF8 deficiency (which shows monocytopenia + absent DCs + granulocytic hyperplasia).
- **Genetic testing:** single-gene *IRF8* sequencing, MSMD gene panels, or WES/WGS; interpret against ClinVar (T80A likely pathogenic, K108E pathogenic).
- **Microbiology/histopathology:** tissue and blood culture with special attention to mycobacteria; granuloma on biopsy (non-caseating or atypical). MSMD can masquerade as

sarcoidosis or Rosai-Dorfman disease until cultures reveal mycobacteria (P 40755768).

- **Differential diagnosis:** other MSMD genes (IL12RB1, IL12B, IFNGR1/2, STAT1, TYK2, SPPL2A, etc.), chronic granulomatous disease, and non-infectious granulomatous/histiocytic disorders.

Treatment

- **Antimycobacterial therapy:** multi-drug regimens tailored to the isolated organism; the partial form is typically **cured** with appropriate antibiotics.
- **Adjunctive IFN-γ:** recombinant IFN-γ (subcutaneous) to bolster macrophage activation in the IL-12/IFN-γ axis (NCIT:C20495 Interferon Gamma).
- **HSCT:** reserved for severe/recessive complete IRF8 deficiency (K108E patient cured by cord-blood transplant, P 25122610); generally **not required** for the partial dominant form.
- **BCG avoidance / management of BCG complications** in known carriers.

Prevention

- **Primary:** avoid live BCG vaccination in individuals with a family history or confirmed *IRF8* variant.
- **Secondary:** cascade genetic testing of relatives; early recognition of unexplained childhood lymphadenopathy with mycobacterial workup.
- **Counseling:** autosomal dominant inheritance implies ~50% transmission risk; genetic counseling and prenatal/preimplantation options where desired.

Other species / model organisms

- **Ortholog:** mouse *Irf8* (HomoloGene group 1629). The **BXH2 mouse** carries an *Irf8* R294 mutation orthologous to human R294/R291Q (P 29128673) and models IRF8 deficiency. Zebrafish and other vertebrates possess DC-like cells with conserved IRF8-dependent programs (P 41379882).

- **Model utility:** dose-dependent knock-in/knockout mice recapitulate DC lineage defects and mycobacterial susceptibility; useful for studying enhancer regulation (P 40680811 P 40378239) and immunotherapeutic IRF8 reprogramming (P 39115195, glioblastoma model).

Evidence Base

PMID	Title (abbrev.)	Role in this report
21524210	*IRF8 mutations and human dendritic-cell immunodeficiency* (Hambleton, NEJM 2011)	**Foundational.** Defines T80A (dominant, partial, selective cDC2 loss, curable BCG disease) vs K108E (recessive, complete). Source of HPO frequencies.
38341181	*830 MSMD patients: systematic review* (Khavandegar 2024)	Epidemiology & clinical frequencies (lymphadenopathy 45.5%, fever 30.2%, organomegaly 24.8%, sepsis 20.8%).
42183200	*MSMD: IFN-γ-driven immunity collapse* (Qian 2026)	22 genes converge on IL-12/23–IFN-γ circuit; ~50% unsolved.
40680811	*Cis/trans regulation of Irf8 enhancers* (Nishiyama & Tamura 2025)	IRF8 dose-dependence of myeloid cell fate — explains subset-selective phenotype.
29128673	*Biallelic IRF8 mutation* (Bigley 2018)	BXH2 mouse ortholog (R294); IRF8 mutants fail to regulate EICE/ISRE.
40072380	*Novel dominant-negative IRF8* (Ham 2025)	Dominant-negative form → decreased cDC2 + mycobacterial susceptibility; broadens phenotype.
25122610	*Functional characterization of IRF8 K108E*	Complete recessive form: absent monocytes/DCs, granulocytic hyperplasia, cured by cord-blood transplant; K108E loses nuclear localization/ stability.
40755768	*Hidden immune defects in childhood granulomatous disorders*	MSMD masquerading as sarcoidosis/ Rosai-Dorfman until cultures reveal mycobacteria — diagnostic caution.
38535546	*Diagnosis & management of infections in MSMD*	BCG vs NTM vs MTB spectrum; culture challenges.
41786143	*IL-12/IFN-γ axis defects and MSMD*	Management review: antibiotics, cytokine therapy, BMT.
39115195	*IRF8 reprogramming in murine glioblastoma*	Confirms IRF8 as master regulator of cDC1 development (mechanistic corroboration).

Evidence source types: human clinical (21524210, 25122610, 29128673, 40072380, 38341181, 40755768), model organism (29128673 BXH2 mouse, 39115195 murine GBM, 41379882 zebrafish), in vitro functional (21524210, 25122610, 40072380, 40680811), database/ computational (gnomAD, ClinVar, UniProt Q02556, EBI OLS4, HPO/JAX, HomoloGene).

Limitations and Knowledge Gaps

1. **Very small case base.** The dominant/partial IRF8 phenotype rests principally on two T80A subjects (Hambleton 2011) plus additional dominant-negative families (Ham 2025). Penetrance, expressivity, sex ratio, and precise prevalence are therefore poorly quantified.
2. **Frequency figures are low-denominator.** HPO frequencies (e.g., BCGitis 2/2, fever 1/2) derive from n=2 and should not be over-interpreted as population estimates.
3. **Direct IL-12/IFN-γ measurements in T80A patients** are sparse; the cytokine-axis steps in the causal chain are partly *inferred* from the shared MSMD circuit rather than demonstrated specifically for T80A.
4. **Genotype–phenotype boundaries** between partial LoF (T80A), dominant-negative (c.1279dupT), and complete LoF (K108E) are still being refined; the dominant-negative form additionally affects pDC/cDC1 and confers viral (EBV/HPV) susceptibility, blurring the classic partial-MSMD picture.
5. **No natural-history or registry data** specific to partial IRF8 deficiency; long-term outcomes, relapse rates, and optimal duration of therapy are undefined.
6. **Modifier genes, epigenetics, and gene–environment interactions** (e.g., BCG strain, mycobacterial burden, endemic TB) are plausible but uncharacterized for this specific genotype.

Proposed Follow-up Experiments / Actions

1. **Aggregate an international IRF8-MSMD cohort** (GeneMatcher, MSMD consortia) to establish penetrance, expressivity, and outcome data for dominant/partial variants specifically.
2. **Deep immunophenotyping** (spectral flow / CyTOF) of T80A carriers to quantify cDC2 vs cDC1 vs pDC vs monocyte fractions and map the exact DC-subset bottleneck.

3. **Functional cytokine assays** (IL-12/IL-23 and IFN-γ production upon TLR/mycobacterial stimulation) in patient PBMCs to convert inferred causal-chain steps 4–6 into demonstrated ones.
4. **Isogenic knock-in models** (T80A heterozygous mouse or human iPSC-derived DCs) to test dose-dependence directly and compare with BXH2/R294.
5. **Single-cell multi-omics of bone-marrow progenitors** to define where in the myeloid hierarchy the partial IRF8 dose becomes limiting.
6. **Prospective trial of adjunctive IFN-γ** vs antimycobacterials alone in genetically confirmed partial IRF8 deficiency to formalize treatment algorithms.
7. **Curate the KB entry** with the ontology appendix below and flag the historical OMIM A/B label swap (IMD32A = 614893 = partial/dominant; IMD32B = 614894 = complete/recessive).

Ontology Term Appendix (for KB population)

Disease: MONDO:0013957 · OMIM:614893 (IMD32A) · Orphanet:319600 · DOID:0111986 · MedGen:814919 · GARD:0017463 **Gene/Protein:** HGNC:5358 *IRF8* · NCBIGene:3394 · Ensembl ENSG00000140968 · UniProt Q02556 · chr16q24.1 **Variants:** NM_002163.4:c.238A>G p.Thr80Ala (ClinVar likely pathogenic) · c.322A>G p.Lys108Glu (ClinVar pathogenic, recessive/IMD32B) **Phenotypes (HP):** HP:0020086 BCGitis · HP:0032252 Granuloma · HP:0002716 Lymphadenopathy · HP:0002840 Lymphadenitis · HP:0001945 Fever · HP:0011463 Childhood onset · HP:0002721 Immunodeficiency · HP:0002719 Recurrent infections · HP:0000006 Autosomal dominant inheritance **Biological processes (GO):** GO:0035722 · GO:0060333 · GO:0097028 · GO:0030099 · GO:0006357 · GO:0042116 · GO:0002250 **Cellular component (GO):** GO:0005634 nucleus **Cell types (CL):** CL:0001057 / CL:0002399 CD1c$^+$/cDC2 · CL:0000990 conventional DC · CL:0000784 pDC · CL:0000576 monocyte · CL:0000235 macrophage · CL:0000775 neutrophil **Anatomy (UBERON):** UBERON:0002371 bone marrow · UBERON:0000178 blood · UBERON:0000029 lymph node · UBERON:0002106 spleen · UBERON:0002107 liver **Infectious agents (NCBI Taxon):** *Mycobacterium bovis* BCG (Taxon:33892) · non-tuberculous mycobacteria · *M. tuberculosis* (Taxon:1773) **Treatments (NCIT):** antimycobacterial antibiotics · NCIT:C20495 Interferon Gamma · hematopoietic stem cell transplantation (severe/recessive only) **Model organism:** *Mus musculus Irf8* (HomoloGene 1629); BXH2 strain (Irf8 R294)

Report compiled across 5 discovery iterations; 7 findings recorded, 18 papers reviewed, ~11 primary PMIDs cited, all identifiers database-verified (gnomAD, ClinVar, UniProt, EBI OLS4, JAX/HPO, HomoloGene).
